

Zhiqian Zhou

Multi-Agent LLM Systems • RAG • Machine Learning • Data Analysis

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Profile

Data Science M.S. student at EPFL; ICML 2026 workshop co-author and two-time 1st-prize hackathon winner. Research interests: LLM multi-agent systems, retrieval-augmented generation (RAG), machine learning and data analysis.

Education

M.S.	EPFL , M.S. in Data Science Focus: Machine Learning and Multimodal Foundation Models	Sep 2026 – Jun 2028 (expected)
B.S.	UPC BarcelonaTech , B.S. in AI GPA 9/10 (Top 3%) Schneider Electric Impact Maker Award 2024. 5 distinctions (Top 5%): HPC, Speech and Dialogue, Knowledge and Reasoning, Modeling and Simulation, Calculus.	Sep 2022 – Jun 2026
Exch.	Beihang University (BUAA) , Exchange Student in AI GPA 96/100 - Courses: Foundation Models, Intelligent Security, Medical Imaging. - HAM10000: ResNet-50 beat DeiT-Small by 0.025 macro-F1 on rare lesions.	Sep 2025 – Jan 2026

Research Experience

AI Research Assistant <i>Sparsity Technologies & DAMA-UPC</i>	Jul – Sep 2025
- Optimized structured-output generation for LLMs under limited compute. - Built a multimodal knowledge graph of flight and rail schedules to model disruption propagation.	Mar 2026 – May 2026
Bachelor's Thesis, UPC (Grade 10/10) <i>HomeCraft: Text-to-Building Generation in a Voxel World</i>	Feb 2026 – Jun 2026
- Built a 6-agent LLM cascade with RAG (2.7K buildings) and 316 executable skills: one prompt to a furnished building. - Benchmarked 5 open-weight LLMs on 50 prompts with an 18-check evaluator validated by 13 humans ($r = 0.84$); best composite 0.86. - Distilled into 2 small Gemma students (QLoRA): no quality gain; identified task completion as the bottleneck.	Advisor: Prof. Ramon Sangüesa

Publication

WinDOM: Self-Family Distillation for Small-Model GUI Grounding. C. Li Chen, Z. Zhou , H. Chen, N. Chauvin. <i>ICML 2026 Second Workshop on Agents in the Wild: Safety, Security, and Beyond</i> . OpenReview arXiv
54K annotation-free GUI grounding corpus; self-family distillation + GRPO lifts a Qwen3.5-2B agent by +5.4 OOD-mean.

Key Projects

AimSafe: Worker Safety Monitoring	2026 1st Prize, MERIThON UPC Repo
- Built the posture and fatigue modules on MediaPipe landmarks: joint-angle unsafe-lift detection, REBA scoring, and drowsiness via adaptive EAR, PERCLOS, microsleep, yawn and gaze. - Part of a real-time (~30 FPS), privacy-preserving system tracking up to 5 workers, fusing risks into a 0–100 safety score with LLM coaching and MQTT machine stops. Python MediaPipe OpenCV NumPy LLM agent MQTT	
Alvidence: Multi-Agent Misinformation Hunter	2025 1st Prize, MERIThON UPC Repo
Co-designed the multi-agent workflow: 3 LLM agents (domain analysis, claim extraction, web-search verification) yielding weighted credibility scores. LangChain Multi-agent Web search Python	
ADDAD: Smadex Creative Intelligence	2026 HackUPC Repo
- Did feature extraction and dataset analysis on 1,076 ad creatives, co-trained a soft-voting ensemble (XGBoost, LightGBM, CatBoost) and analyzed results: test macro-F1 0.677, AUC 0.94/0.98 for top/under-performers, ECE 0.096. - Team also fine-tuned SmolVLM (JSON critiques) and Flux LoRA/DPO (visual rebuilds). XGBoost LightGBM CatBoost SmolVLM	
NextBuy: Smart Demand Signals for Inibsa	2026 3rd Prize, INTERHACK BCN Repo
Did feature engineering and data analysis on clinic sales history; co-trained 90-day purchase/churn models (temporal holdout: LightGBM AUC 0.87, P@5% 0.95; XGBoost AUC 0.81) and analyzed SHAP drivers. LightGBM XGBoost SHAP Gemini FastAPI	
CUPME: Enterprise RAG Chatbot for Siemens Energy	2024 Repo
Built the RAG architecture (bge-small embeddings in Pinecone, Gemini 1.5 Flash) and the backend–frontend interaction, including chat-history management; answers in 4 languages via LLM query translation. LangChain Pinecone RAG FastAPI	

Skills & Activities

Programming: Python (Advanced), C++, SQL, R, MATLAB ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
GenAI & Agents: LangChain, RAG, ReAct, MCP, Tool Use Vision & NLP: CNNs, ViT, OpenCV, MediaPipe
MLOps & Tools: Git, Docker, FastAPI, Weights & Biases, Jira Languages: Chinese (Native), Spanish/Catalan (Bilingual), English (Professional)
Leadership: AECH UPC events committee (2023–2026): 100+ attendee networking events; hosted a Tsinghua delegation
Interests: Boxing, dancing, travel, gastronomy